

Evaluation of Housing and Management Conditions Affecting Animal Welfare in Dairy Cattle Farms in Ankara Province, Türkiye

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Abstract: Shelter conditions and herd management practices significantly influence the optimization of production performance in dairy cattle when assessing animal welfare. This study was conducted to evaluate the effects of housing structure and dimensions, climatic conditions, calf rearing practices, milking hygiene, and herd health practices on animal welfare in dairy cattle farms in the some of town in Ankara.

Information regarding shelter design, bedding use, calf care and feeding, barn cleaning and animal health status, herd management, and milking practices in the farms subject to evaluation was obtained through on-site observation, questionnaires, and interviews with staff. Statistical analysis showed that the barn structure and climatic conditions in the farms were within acceptable limits, 80% of the farms used bedding, 60% of the animals were dirty and 13.33% were very dirty in the contamination scoring, 3.05% of diseases were caused by mastitis, and the lameness rate was 85.67%. Also, average of the carbon dioxide (CO₂) level in shelters was 1049.70 ppm, which is within acceptable values. Furthermore, it has been evidenced that staff training is one of the fundamental principles underpinning animal welfare practices.

In conclusion, this study exhibits that improving animal welfare in dairy cattle management is achievable through enhanced shelter and environmental conditions, improved calf rearing practices, milking hygiene, protective approaches to herd health, and enhanced staff training.

Keywords: Stress, behaviour, cows, environment, shelter

Ankara İli Süt Sığırı Çiftliklerinde Hayvan Refahını Etkileyen Barınma ve Yönetim Koşullarının Değerlendirilmesi

Öz: Hayvan refahı değerlendirilirken, barınak koşulları ve sürü yönetimi uygulamaları, süt sığırlarında üretim performansının optimizasyonunu önemli ölçüde etkiler. Bu çalışma, Ankara'nın bazı ilçelerindeki süt sığırı çiftliklerinde barınak yapısı ve boyutları, iklim koşulları, buzağı yetiştirme uygulamaları, sağım hijyeni ve sürü sağlığı uygulamalarının hayvan refahı üzerindeki etkilerini değerlendirmek amacıyla yapılmıştır. Değerlendirmeye tabi tutulan çiftliklerdeki barınak tasarımı, altlık kullanımı, buzağı bakımı ve beslenmesi, ahır temizliği ve hayvan sağlığı durumu, sürü yönetimi ve sağım uygulamalarına ilişkin bilgiler, yerinde gözlem, anketler ve personel ile yapılan görüşmeler yoluyla elde edilmiştir. İstatistiksel analiz, çiftliklerdeki ahır yapısı ve iklim koşullarının kabul edilebilir sınırlar içinde olduğunu, çiftliklerin %80'inin altlık kullandığını, hayvanların %60'ının kirliliği ve %13.33'ünün çok kirliliği olduğunu, hastalıkların %3.05'inin mastitis kaynaklı olduğunu ve topallık oranının %85.67 olduğunu göstermiştir. Ayrıca, barınaklardaki karbondioksit (CO₂) seviyesinin ortalaması 1049.70 ppm olup, kabul edilebilir değerler içindedir. Dahası, personel eğitiminin hayvan refahı uygulamalarının temel ilkelerinden biri olduğu kanıtlanmıştır. Sonuç olarak, bu çalışma, süt sığırı yönetiminde hayvan refahının iyileştirilmesinin, barınak ve çevre koşullarının iyileştirilmesi, buzağı yetiştirme uygulamalarının geliştirilmesi, sağım hijyeni, sürü sağlığına yönelik koruyucu yaklaşımlar ve personel eğitiminin geliştirilmesi yoluyla mümkün olduğunu göstermektedir.

Anahtar Kelimeler: Stres, davranış, inek, çevre, barınak

INTRODUCTION

The world population, industrialization, and per capita income are showing an increasing trend (Thornton, 2010). According to United Nations data, the world population is expected to reach 9.8 billion by 2050 (UN, 2022). Consequently, demand for animal products is also increasing along with population growth. In order to meet this growing demand, industrial animal farming systems have been adopted, where animal husbandry is more intensive and increasing yield per animal is prioritized (Neethirajan, 2023). However, the proliferation of these production models, the increase in the number of animals raised on farms, environmental sustainability, health and animal welfare, social attitudes towards animals, and the tests and scientific

studies conducted on animals have led to growing interest and concern for animal welfare (Etim et al., 2014).

Inappropriate housing and feeding conditions, diseases, weaning, overcrowding, isolation, and climatic factors are among the main sources of stress that negatively affect animal welfare (Veissier et al., 2008). When stressed, animals use their energy and productivity processes to cope with stress, which leads to productivity and health losses. Therefore, improving environmental conditions in parallel with the genetic development of animals is becoming increasingly important in modern livestock systems (Van

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Hertem et al., 2016). The temperature-humidity index (THI) is a widely used indicator for assessment heat stress in animals; it is influenced from environmental conditions, shelter structural characteristics, facility orientation and roof isolation properties (Lovarelli et al., 2021). The physiological status of animals shapes the extent of their response to the conditions they are exposed to. Periods such as late pregnancy, calving, and the beginning of lactation are among the most physiologically sensitive periods for animals. Exposure to heat stress during these periods, when animals are highly sensitive to their environment, has negative effects on milk yield, feed intake, rumination, and reproductive performance. In fact, it has been reported that long-term heat stress causes a decrease in thyroxine, growth hormone, glucocorticoid concentrations and basal metabolism in cattle (Collier et al., 1982). High-yielding cows, particularly when managed under high THI conditions, display more pronounced physiological stress responses, suggesting that high-yield animals are more sensitive to heat stress (Valdivia-Cruz et al., 2021).

As the dairy cattle industry moves toward more concentrated and confined production models, the transition from pasture-based systems to free-stall or tie-stall shelter systems has become widespread. This change often restricts freedom for voluntary movement and behavioural expression, leading to an increase in hoof and leg problems, a reduction in lying time, and an increase in abnormal lying behaviour (Shepley et al. 2019; Shepley and Vasseur, 2021). Similarly, poor housing hygiene is also considered a stress factor and negatively affects foot-hoof health and udder health (Ruud et al., 2010; Munoz et al., 2008; Breen et al., 2009; Sant'Anna and Da Costa, 2011). Additionally, dirty and wet coats in animals reduce thermoregulation ability, thereby increasing stress levels (Algers et al., 2009). Cattle, which in their natural environment can experience different sensory experiences such as socializing in complex groups, exploring, consuming various feed materials, and seeking shelter, do not have the opportunity to exhibit such behaviours under intensive conditions (Zobel et al. 2019; Rørvang et al. 2018). Therefore, environmental enrichment such as mechanical brushes or various objects to play positively affect animal welfare by providing behavioural diversity (DeVries et al., 2007; Bøe et al., 2012). From this perspective, shelters conditions significantly affect the optimization of production performance in dairy cattle farming, including small-scale family farms (Prabowo et al. 2025).

While many studies have explored how shelters and rearing environments influence welfare in dairy cattle, the majority of this research has been carried out either in other climate zones or large-scale farms. In Turkey, field data on small and medium-scale dairy cattle farms, particularly in semi-arid climate zones, are limited. Therefore, this study aims to evaluate shelter characteristics, husbandry practices and some of welfare-related animal indicators in dairy cattle farms located in Ankara province.

MATERIAL AND METHODS

General Information About the Businesses

This study was performed in September and October 2019 in a total of 15 dairy cattle farms located in the Ayaş, Sincan, Polatlı, Temelli, Çubuk, and Kazan town of Ankara province. The farms were selected to represent the region by having different shelter types (closed and semi-open) and herd size.

Data Collection and Evaluation Criteria

The data collection form used in the study was prepared based on the Welfare Quality Assessment Protocol for Cattle, which is widely used in animal welfare assessments, and adapted to field conditions. The form includes information such as barn type, shelter dimensions, floor structure, bedding use, environmental enrichment practices, care and feeding practices, and herd management.

Environmental Measurements

In shelters temperature (°C) and relative humidity (%) measurements were taken using a Tt-Technic HTC-1 digital thermometer-hygrometer. CO₂ concentrations were measured using a Green Eye/AZ7798 datalogger. Measurements were taken in the each shelters, approximately 1.5 m above the ground, during daylight hours, and the averages of the obtained values were used in the analyses.

Animal Welfare Indicators

The cleanliness situation of the animals was evaluated in three categories: clean, dirty, and very dirty, considering the udder, the area between the navel and udder, and the hind legs. This evaluation was based on previously reported dirtiness scoring methods (Ruud et al., 2010). Additionally, the number of lame animals in the herd, the presence of superficial wounds, and disease status were recorded.

Farm Management and Staff Information

Information on care and feeding practices, herd health and management, and the training situation of employees in cattle breeding or animal welfare was obtained through interviews with farm owners and employees.

Statistical Analysis

The obtained data were analyzed using the IBM SPSS Statistics v.23 software package. Descriptive statistics were presented by calculating the percentages (%), means, minimum, maximum, and standard deviations of the parameters examined.

RESULTS AND DISCUSSION

General Characteristics of Farms

In this research, 15 dairy cattle farms were evaluated. It was determined that 60% of the examined farms had a closed barn system, while 40% had a semi-open barn system (Table 3.1). The main cattle breeds raised in the farms were Brown Swiss (Montofon), Simmental, and Holstein Friesian. The total number of cattle on the farms is 3,389 of which 1,765 are cows. The average number of cattle per farm was

calculated as 226. The average number of cows, heifers, steers, and calves per farm was 118, 60, 16, and 32, respectively (Table 1).

Table 1. Animal information at the farms

Variables	$\bar{x}$	Min	Max	S
Number of dairy cattle farms	15			
Shelter type (%)	Kapalı %60		Yarı Açık %40	
Cattle breeds on the farm	Montofon (Brown Swiss), Simental, Holstein			
Cow	118	10	350	98,71
Heifer	60	13	180	52,98
Bull	16	1	90	22,84
Calf	32	2	105	32,07
Total number of cattle on the farm (head)	3389			
Total number of cow on the farm (head)	1765			
Total number of heifer on the farm (head)	226	31	560	181,85

Farm structure and herd composition are among the fundamental elements that must be considered when evaluating animal welfare. Having a balanced structure of animals of all ages in the herd is important for sustainability in production and animal welfare. In Turkey, housing planning and husbandry practices in small and medium-sized operations are often not in line with welfare assessment criteria (Sert and Uzmay, 2017).

Housing and Rearing Conditions

The shelter measurements and findings related to care and feeding for the studied farms are presented in Table 3.2. Stalls were found in all farms, with an average stall width of 110.33 cm and an average stall length of 216.33 cm. It was determined that 80% of the farms used bedding. It was determined that the vast majority of barn floors were concrete (93.33%), while 6.67% had dirt floors. All farms evaluated provided dedicated birthing pens, and most (80%) also maintained a separate isolation pen for sick animals. Fluorescent lighting constituted the predominant system across shelters (86.67%), whereas a smaller proportion (13.33%) utilized incandescent bulbs (Table 2).

Shelter conditions such as floor structure and bedding use have a direct effect on animals' resting period and lying behavior, making them important for animal health and welfare. Bedding material was not used in 20% of the farms evaluated. Whereas, researches evaluating the effects of different bedding materials on animal welfare report that appropriate bedding use advances both animal comfort and welfare indicators (Suárez-Usbeck et al., 2023; Eberl et al., 2024). Besides, the conformity of barn measures to the animal's body structure, the space per animal and stock density are also decisive factors for welfare. High stock density restricts feed intake, lead to a reduce in milk yield and increased competition among animals (Bach et al., 2008). Therefore, the management of indoor space must be evaluated not only in terms of structural sizes but also in conjunction with stock density.

Calving is a physiological process associated with high levels of stress for animals, and insufficient hygiene conditions during this process can negatively affect calf and cow health. Robertson et al. (2022) reported that high stocking density,

inadequate bedding use, and cleaning practices in birthing pens make it difficult to raise healthy and resilient dairy cows in the long term. The number and sizes of feed trough and drinking buks are also factors that can affect animals' stress levels in terms of feed intake and social hierarchy. As reported by the National Research Council (2001), the average enough feed through space per cow is 47 cm, and the average sufficient water space per cow is 6 cm (Welfare Quality, 2025). In farms evaluated according to these advices, the average feed trough width was found to be above the ideal value. Lighting type and time in the shelter is also among the environmental factors that effect animal welfare and performance. The literature notices that proper lighting conditions are associated with calf growth, milk yield, reproductive performance, and general animal health (Adamczyk et al., 2024). Indeed, studies examining lighting programs on calf growth and development have found that lighting programs have an effect on animal performance in later periods (Penev et al., 2014; Patbandha et al., 2016). Additionally, it has been reported that long-term lighting practices with appropriate light intensity can have positive effects on milk yield and cortisol concentrations (Lim et al., 2021; Son et al., 2020).

Climatic Factors and Air Quality

Indoor temperature, humidity, and CO₂ measurements from the shelters surveyed are summarized in Table 3. The average indoor temperature, relative humidity, and carbon dioxide (CO₂) levels were found to be 19.74 °C, 66.50%, and 1049.70 ppm, respectively.

The temperature-humidity index (THI), which allows for the combined assessment of temperature and humidity, is a widely used indicator for determining heat stress in cattle (Valdivia-Cruz et al., 2021). Therefore, THI has been found to be within the optimal range for cattle in the evaluated farms. On the other hand, the literature notices that increases in THI values are associated with decreased feed intake, lessen milk yield, and increased physiological stress responses (Valdivia-Cruz et al., 2021). Similarly, Bokharaeian et al. (2023) has determined that seasonal and monthly changes in THI remarkably affected milk yield and quality, and that high THI values increased somatic cell count and total

bacterial count. Also, CO₂ itself is not highly toxic to cattle, but its build-up signals poor ventilation and insufficient fan operation. Accordingly, it has been reported that keeping

CO₂ concentrations below 3,000 ppm in dairy cattle shelters is acceptable in terms of air quality (CIGR, 1984).

Table 2. Shelter dimensions, equipment, maintenance and feeding values

Variables	$\bar{x}$	Min	Max	S
Door height (m)	3.87	2	5.5	0.92
Door width (m)	4.44	1.5	6	1.12
Stall width (cm)	110.33	90	125	11.87
Stall length (cm)	216.33	170	285	31.92
Use of bedding (%)	80	-	-	-
Barn floor – concrete (%)	93.33	-	-	-
Barn floor – soil (%)	6.67	-	-	-
Calving pen (%)	100	-	-	-
Sick pen (%)	80	-	-	-
Exercise area (m ² /head)	105	-	-	-
Lighting – incandescent bulb (%)	13.33	-	-	-
Lighting – fluorescent (%)	86.67	-	-	-
Feed bunk width (cm)	64.60	50	80	8.51
Feed bunk height (cm)	34.00	15	40	6.49

Therefore, the average CO₂ concentrations we obtained from the farms are below this value. In addition, temperature and humidity on farms can influence ventilation performance.

Table 3. Indoor temperature, humidity, and carbon dioxide values

Variables	$\bar{x}$	Min	Max	S
Temperature (°C)	19.74	15	24	2.98
Relative humidity (%)	66.50	56	76	6.25
Carbon dioxide (CO ₂ , ppm)	1049.70	100.00	4700.00	764.98

Calf Management and Rearing Practices

Calf sheltering, feeding routines, and daily care practices across the farms are summarized in Table 4. Nearly half of the calves were kept in calf hutches (46.66%), followed by closed individual pens (33.34%) and closed group pens (20%). Dehorning was uncommon, applied on calves in only 13.34% of the farms. Calves were weaned at an average of 88.34 days. Mean space allowance per calf was 4.96 m², with recorded values ranging from 0.96 to 26.30 m².

Table 4. Values related to shelter type, care, and feeding practices in calves

Variables	$\bar{x}$	Min	Max	S
Calf housing – group (%)	20	-	-	-
Calf housing – individual (%)	33.34	-	-	-
Calf housing – calf hutch (%)	46.66	--	-	-
Weaning age (days)	88.34	65	160	28.19
Space allowance for calves (m ² /head)	4.96	0.96	26.30	4.86

Calf care and feeding practices constitute one of the most critical periods in dairy cattle operations, laying the foundation for animal welfare. This investigation noted that 33.34% of the farms reared calves in individually-enclosed, while 20% utilized enclosed group pens. The type of stereotypical behavior observed in calves is directly related to shelter conditions. Behavioral disorders such as sucking, licking, biting objects, tongue rolling, and head shaking are frequently seen in calves housed in individual pens. In group pens where calves are in contact with each other, social stereotypies involving sucking each other and manipulating other individuals are observed. The frequency of these behaviors rises, particularly during the pre-weaning nutrition period (Iwona, 2021). In addition, when calves are kept in individual or closed group shelters, opportunities for stable social bonding become restricted, which may disturb herd social structure and delay the formation of long-term social relationships (Pilarczyk et al., 2025). In this investigation, calves are separated from their mothers immediately after birth and calves were fed milk for an average of 88.34 days. If calves are allowed to suckle their mothers, the relationship between the mother and calf strengthens, making the subsequent separation process remarkably stressful and hindering the calves' ability to friendship with other calves (Fröberg and Lidfors, 2009; Roth et al., 2009).

Milking, Herd Health, and Animal Hygiene Scores

The study revealed that the usage rates of central milking systems and mobile milking machines in the examined farms were 93.33% and 6.67%, respectively. Milking was routinely carried out twice per day in all shelters, with udder cleaning and pre-milking hygiene checks applied as standard practice. Disease incidence rates in the farms were calculated as 3.05%, 1.31%, 2.47%, 1.40%, and 2.28%, respectively, for

mastitis, hypocalcemia, ketosis, acidosis, and laminitis. Furthermore, the average milk yield in the farms was determined to be 27.13 kg/day, varying between 16 and 37 kg/day.

When the cleanliness situation of the cattle was evaluated, it was determined that 26.66% of the animals were clean,

60.00% were dirty, and 13.33% were very dirty. Furthermore, it was determined that 40% of the staff in the farms had received training in cattle farming, while 6.67% had received training in animal welfare (Table 5).

Table 5. Values related to milking, diseases, and hygiene scores in farms

Variables	$\bar{x}$	Min	Max	S
Milking method – mobile milking machine (%)	6.67	-	-	-
Milking method – centralized milking system (%)	93.33	-	-	-
Mastitis (%)	3.05	0.77	10.71	2.92
Hypocalcemia (%)	1.31	0	3.77	1.25
Ketosis (%)	2.47	0.22	6.67	2.17
Acidosis (%)	1.40	0.34	3.22	0.76
Laminitis (%)	2.28	1.02	4.90	1.10
Milk yield (kg/day)	27.13	16	37	5.60
Hygiene Scores – clean (%)	26.66	-	-	-
Hygiene Scores – dirty (%)	60.00	-	-	-
Hygiene Scores – very dirty (%)	13.33	-	-	-

Milking technique and practices related to milking hygiene are important management factors in remaining udder health and animal welfare in dairy cattle. The findings observed in this study concerning lameness and udder health indicate that herd health problems are not solely a clinical matter but also a multidimensional risk factor that simultaneously effects welfare and production. Also, it has been reported that lameness and mastitis, either alone or in combination, have considerable negative effects on milk yield in literature (Prasomsri, 2022).

A key indicator of the quality of the environment in which cattle are sheltered is how clean or hygienic the cows are. Indeed, raised uncleanness among cattle leads to udder infections and productivity losses (M'hamdi et al. 2012; Belay, 2024). It can be concluded that the hygiene score of animals in the evaluated farms negatively affects their welfare. In addition, changes made in practice to improve animal welfare generally require additional training for staff (Fernandes et al. 2021). Indeed, training programs targeting the basic attitudes and behaviors of animal breeders have been found to reduce fear and increase productivity in dairy cattle and pigs (Coleman et al. 2000; Hemsworth et al. 2002).

CONCLUSION

This study evaluated the effects of housing structure, microclimatic conditions, calf rearing practices, milking hygiene, and herd health management on animal welfare in dairy cattle farms in Ankara province based on field data. The results obtained display that animal welfare is not linked on a single factor; shelter conditions, environmental factors, and management practices must be considered as a whole. Shelter type, bedding use, and walking area appear as crucial factors in terms of animals' resting behavior and comfort. The investigation reported in this paper microclimatic

conditions were generally within acceptable limits, the effects of ventilation performance and within-day environmental variability on welfare should not be overlooked. Early intervention practices such as weaning, separation from the mother, and housing type have long-term behavioral and physiological effects on welfare.

Similarly, milking technique, herd health, and hygiene status have been evaluated as management factors directly concerned to animal welfare. It is known that health problems such as mastitis and lameness not only cause production losses but also lead to welfare impairments associated with pain and stress in animals. Staff training is among the key factors determining the effectiveness of milking hygiene and herd health practices.

In conclusion, this study demonstrates that animal welfare outcomes in dairy farms can be strengthened by better-designed shelters, a more controlled and welfare-oriented farm microclimate, structured calf-rearing routines, and regular staff training that becomes a standard operational requirement.

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